

Chapter 13

CS2 course package - University of Southern Indiana (USI)

Srishti Srivastava

University of Southern Indiana

Abstract

The ubiquitous nature of parallel and distributed computing (PDC) in modern software systems requires an early and deliberate introduction to these concepts within the undergraduate computer science curriculum. While foundational tracks historically isolate students within a strict sequential programming paradigm, this chapter presents a successful course exemplar from the University of Southern Indiana (USI) that modernizes the CS2 course, “Intermediate Object-Oriented Programming”, by integrating PDC principles directly into core topics. This curriculum redesign incrementally introduces students to concurrent thinking through a balanced mix of unplugged activities and plugged-in programming modules. As students master intermediate Java concepts, they simultaneously explore data parallelism, event-driven programming, and parallel processing via Java multithreading.

This chapter outlines the pedagogical motivations, architectural course changes, and design decisions behind the transformation. It provides a detailed look at how active learning strategies and hands-on, project-based assignments allow students to absorb complex, concurrent concepts in an accessible, highly contextualized environment. Moreover, insights on student engagement, concept retention, and practical programming competency, offering actionable recommendations are provided for educators looking to foster a parallel computing mindset in early CS courses.

Descriptor Elements

Programming Language Employed: Java

Relevant core courses: CS2

Pre-requisites: At USI, students enrolling in the CS2 course should have completed their CS1 course with a minimum of D grade or higher. They should have also completed a discrete math course with a minimum of D grade or higher.

Relevant PDC topics: |List topics here with bloom classification (use K-knowledge, C-comprehension, or A-be able to apply)|

Relevant PDC topics and Blooms levels

PDC Concept	Chapter Section				
	1.1	1.2	1.3	1.4	1.5
Topic1	K	C	A		
Topic2	K	C	A		
Topic3		C	A	A	

Learning outcomes:

1. |Outcome 1|
2. |Outcome 2|

Context for use: |Your local context about institution, department, mix and levels of students, class size, language employed, etc.|

13.1 Introduction

|Motivation and background for the chapter.|

13.2 How CS2 was changed and PDC topics integrated

|Explain how the existing course (e.g., CS1) was modified.|

13.2.1 Integrated Syllabi (Pre and Post)

|Provide high level syllabi highlighting changes.|

13.2.2 Highlights

|What remained the same, and what specifically needed to be changed or deleted?|

13.2.3 Challenges

|Discuss logistical, pedagogical, or technical difficulties.|

13.3 New Unplugged Activities

13.3.1 The Overly Complicated Machine

Problem Description, Prerequisites, and Learning Outcomes

This activity introduces the concept of event handling and event based programming to students.

Implementation Details

- Logistics (Material required for the activity, etc.)
- How to Conduct
- Instructional materiel (Lectures, Videos, Assignments, etc.)

Experience and Teaching Tips

Data and Analysis

13.3.2 Fork Join on Post-its

Problem Description, Prerequisites, and Learning Outcomes

This activity introduces the concept of data parallelism, multithreading, and concurrent computing.

Implementation Details

- Logistics (Material required for the activity, etc.)
- How to Conduct
- Instructional material (Lectures, Videos, Assignments, etc.)

Experience and Teaching Tips

Data and Analysis

13.4 New Plugged-in Activities

13.4.1 Fork-Join Parallel Sum

Problem Description, Prerequisites, and Learning Outcomes

This activity introduces the concept of data parallelism, multithreading, and concurrent computing.

Algorithms

Implementation Details

- Instructional Material (Lectures, Videos, Assignments, Tests)
- How-To Guides

Experience and Teaching Tips

Data and Analysis

13.4.2 Seeing Concurrency: A Java-Based Parallel Image Processing

Problem Description, Prerequisites, and Learning Outcomes

This activity introduces the concept of data parallelism, multithreading, and parallel computing.

Algorithms

Implementation Details

- Instructional Material (Lectures, Videos, Assignments, Tests)
- How-To Guides

Experience and Teaching Tips

Data and Analysis

13.5 Course-wide Pre and Post Course Surveys

To evaluate the impact of integrating PDC concepts into the intermediate computer science curriculum, a comparative data collection study was conducted across two consecutive Spring semesters in the CS2 course. Baseline data was collected in Spring 2025 for a traditional CS2 curriculum structure focused strictly on sequential object-oriented programming without any explicit PDC interventions. To assess student growth and conceptual shifts, identical pre- and post-test surveys were administered during the first and final weeks of the semester. The same pre- and post-survey methodology was employed in Spring 2026 to measure the efficacy of a targeted pedagogical intervention. This intervention was structurally delivered through the two unplugged exercises and the two plugged-in coding modules discussed above.

By maintaining Java as the instructional language across both cohorts, the study isolated the curriculum redesign as the primary variable. The comparative results strongly indicate that the active-learning PDC intervention effectively improved CS2 students' self-reported understanding and confidence in PDC concepts.

13.5.1 Results and Analysis : Spring 2025 Survey Data

13.5.2 Results and Analysis : Spring 2026 Survey Data (PDC Intervention)

13.5.3 Did the PDC Intervention Improve Learning?

13.6 Other Activities and Ideas

[Alternative ways to apply these concepts, etc..]

13.7 Conclusion

Bibliography

Appendix

|You must include these four specific subsections (in addition to anything else):|

Github Link to Instructional Material

`https://github.com/...`

Detailed Pre and Post Syllabus

Organization of Material

|Explain the folder structure of your Github repository.|

Survey and Other Anonymized Data and Analysis

|Include student feedback, learning data, or implementation analysis.|